

UNIVERSITY OF KENTUCKY

ISC 450G: Applied Artificial Intelligence Literacy

Term: Fall 2026

Credit Hours: 3.0

Format: Asynchronous Online & Synchronous Seminars (Hosted via Canvas)

Part I: Instructor and Course Information

Instructor Contact

- **Instructor:** [Your Name/Title]
- **Office Location:** [Your Office] / Zoom Room [Link]
- **Office Hours:** Tuesdays 1:00 PM – 3:00 PM, or by appointment.
- **Email:** [Your Email]@uky.edu
- **Communication Policy:** I check Canvas Inbox and email twice daily on weekdays. Please allow up to 24 hours for a response Monday–Friday, and 48 hours over the weekend. For complex questions about architecture or project scoping, please schedule a Zoom appointment or visit office hours.

Course Description

This interdisciplinary course bridges the gap between theoretical computer science and practical, everyday applications of Artificial Intelligence (AI). Designed for both undergraduate and graduate students, ISC 450G demystifies generative models, neural networks, and algorithmic decision-making. Students will transition from passive consumers of AI to critical, proficient operators. The curriculum heavily emphasizes hands-on prompt engineering, the ethical deployment of AI in critical sectors (such as healthcare, education, and logistics), and the socioeconomic impacts of automation.

Student Learning Outcomes (SLOs)

Upon successful completion of this course, students will be able to:

1. **Explain** the underlying mechanics of Large Language Models (LLMs) and machine learning algorithms without relying on anthropomorphism or technical jargon.
2. **Design and execute** advanced prompt engineering strategies to solve complex, domain-specific problems using current generative AI tools (e.g., ChatGPT-4o, Claude).

3. **Critically evaluate** AI outputs for hallucinations, algorithmic bias, and logical inconsistencies.
4. **Analyze** the ethical, legal, and privacy implications of deploying AI in high-stakes environments.
5. *(Graduate Students Only)* **Synthesize** primary research to propose novel AI implementations or policy frameworks within a specific professional discipline.

Required Materials and Technologies

- **Primary Platform:** Canvas (All modules, assignment submissions, and readings are hosted here).
- **Required Accounts:** Students must create free accounts for at least two LLMs (e.g., OpenAI's ChatGPT, Anthropic's Claude, or Google's Gemini).
- **Textbook:** *Artificial Intelligence: A Guide for Thinking Humans* by Melanie Mitchell (Free digital access via UK Libraries).
- **Supplemental Readings:** Provided weekly via Canvas (e.g., white papers, peer-reviewed medical and technical journals, task force reports).

Part II: Grading and Evaluation

Grade Breakdown

Assessment Category	Undergraduate Weight	Graduate Weight (400G)
Weekly Canvas Discussions	20%	15%
Quizzes (4 total)	20%	15%
Hands-On AI Case Studies (3)	30%	20%
Midterm Examination	15%	15%
Final Applied Project	15%	20%
Graduate Seminar Leadership	N/A	15%

400G-Level Distinction

Graduate students enrolled in the 400G level are held to higher analytical standards. They are required to lead one synchronous seminar discussion based on advanced literature and must complete an expanded final project (4,500 words minimum) that includes original data analysis or a comprehensive systems architecture proposal.

Undergraduate Grading Scale: A (90-100%), B (80-89%), C (70-79%), D (60-69%), E (Below 60%).

Graduate Grading Scale: A (90-100%), B (80-89%), C (70-79%), E (Below 70%). *Note: The Graduate School does not award 'D' grades.*

Part III: Course Policies

Generative AI Acceptable Use Policy

Because this is an AI literacy course, the use of Generative AI is not only allowed but strictly required for designated assignments. However, academic integrity remains paramount.

- **Allowed Use:** Brainstorming, outlining, debugging code, and generating initial drafts *for specific assignments where indicated.*
- **Prohibited Use:** Using AI to complete multiple-choice quizzes, substituting AI generation for peer-review feedback, or submitting unedited AI outputs as your final work without fact-checking.
- **Citation Requirement:** Any time an AI tool is used to generate text, code, or ideas, you must include a "Methodology Appendix" at the end of your assignment detailing which tool was used, the exact prompts entered, and how you verified the output's accuracy. Failure to provide this documentation will result in a zero for the assignment and a potential academic offense report.

Academic Integrity

Per Senate Rule 6.3.1, students shall not plagiarize, cheat, or falsify academic records. All academic work, written or otherwise, submitted by students to their instructors or other academic supervisors, is expected to be the result of their own thought, research, or self-expression.

Accommodations Due to Disability

If you have a documented disability that requires academic accommodations, please consult with the Disability Resource Center (DRC) immediately. The DRC will provide a Letter of Accommodation. Contact: drc@uky.edu or (859) 257-2754.

Diversity, Equity, and Inclusion

The University of Kentucky is committed to our core values of diversity and inclusion, mutual respect and human dignity, and a sense of community. This classroom will be a safe, inclusive environment where diverse perspectives are welcomed and respected.

Part IV: Detailed Weekly Schedule

Note: Readings and assignments are to be completed by Friday at 11:59 PM EST of the respective week, unless otherwise noted in Canvas.

Module 1: Foundations of Artificial Intelligence

Week 1 (Aug 24 – 28): The Hype vs. The Reality

- **Lecture:** What is an algorithm? A brief history from symbolic AI to neural networks.
- **Readings:** Mitchell, Chapter 1; "Stochastic Parrots" (Bender et al., Excerpt).
- **Assignment:** Canvas Discussion Board - Introduce yourself and share an example of an AI tool you interact with daily.
- **Assessment:** Syllabus & AI Policy Quiz.

Week 2 (Aug 31 – Sep 4): Under the Hood of Large Language Models

- **Lecture:** How Transformers work. Understanding tokens, context windows, and training parameters.
- **Readings:** Mitchell, Chapters 2-3.
- **Assignment:** Concept mapping exercise—visually map the flow of data in an LLM from input to output.

Week 3 (Sep 7 – 11): Advanced Prompt Engineering

- **Lecture:** Moving beyond simple queries. Few-shot prompting, chain-of-thought, and assigning personas.
- **Readings:** OpenAI Prompt Engineering Guide.
- **Assignment: Case Study 1 (Due):** Use an LLM to simplify a dense academic abstract. Submit your iterative prompt history and the final output.

Module 2: Ethics, Bias, and Algorithmic Harm

Week 4 (Sep 14 – 18): Algorithmic Bias and Training Data

- **Lecture:** How human biases are encoded into machine learning models (e.g., facial recognition, hiring algorithms).
- **Readings:** Mitchell, Chapter 4; ProPublica's "Machine Bias" report.
- **Assessment:** Quiz 1 (Foundations and Prompting).

Week 5 (Sep 21 – 25): Copyright, Creativity, and Deepfakes

- **Lecture:** Generative imagery, synthetic media, and the crisis of verifiable truth.
- **Readings:** Canvas selections on the US Copyright Office's stance on AI art.
- **Assignment:** Canvas Discussion - The ethics of deepfakes in political campaigning.

Week 6 (Sep 28 – Oct 2): Privacy, Data Scraping, and Regulation

- **Lecture:** Global regulatory frameworks including the EU AI Act and state-level task force guidelines.
- **Readings:** Blueprint for an AI Bill of Rights (OSTP).
- **Assessment: Case Study 2 (Due):** Algorithmic Auditing. Students will run a series of prompts to intentionally trigger and document a model's safety guardrails.

Module 3: AI in Critical Sectors (Midterm)

Week 7 (Oct 5 – 9): Midterm Evaluation

- **Lecture:** Midterm review and synthesis.
- **Assessment: Midterm Examination** (Timed via Canvas. Covers Modules 1 & 2).

Week 8 (Oct 12 – 16): AI in Software and Systems Architecture

- **Lecture:** AI coding assistants (VS Code Copilot, Claude), automating QA, and accelerating development cycles.
- **Readings:** Selected articles on AI-assisted software engineering.
- **Assessment:** Quiz 2 (Ethics and Regulation).

Week 9 (Oct 19 – 23): Clinical and Healthcare Applications

- **Lecture:** Evaluating LLM performance in medical contexts. Automating clinical documentation and diagnostic support.

- **Readings:** Peer-reviewed study: *Evaluating ChatGPT-4o's performance in grading medical student SOAP notes*.
- **Assignment:** Canvas Discussion - The implications of AI handling patient data.

Week 10 (Oct 26 – 30): Logistics and Autonomous Systems [Fall Break Oct 26-27]

- **Lecture:** Predictive logistics and autonomous navigation.
- **Readings:** Canvas module on drone navigation algorithms and payload logistics.
- **Assessment: Case Study 3 (Due):** Analyze the systemic requirements and edge-cases for using drone networks to deliver whole blood to trauma patients in rural Kentucky.

Module 4: The Future of Automation and Final Projects

Week 11 (Nov 2 – 6): The Future of Work

- **Lecture:** Job displacement, AI as a collaborative "co-pilot," and workforce upskilling.
- **Readings:** Mitchell, Chapter 5.
- **Assessment:** Quiz 3 (Applied AI).

Week 12 (Nov 9 – 13): Graduate Seminars

- **Lecture:** Live Zoom seminars led by 400G students.
- **Assignment:** Undergraduate students must submit peer-evaluations of graduate presentations.

Week 13 (Nov 16 – 20): Institutional AI Policies

- **Lecture:** How universities and hospitals build AI guidelines (e.g., AI@UK task force initiatives).
- **Assignment:** Submit a one-page project proposal for the Final Applied Project.

Week 14 (Nov 23 – 27): Thanksgiving Break

- *No classes. Independent work on final projects.*

Week 15 (Nov 30 – Dec 4): Project Consultations

- **Lecture:** Open office hours and 1-on-1 final project troubleshooting.
- **Assessment:** Quiz 4 (Course Synthesis).

Week 16 (Dec 7 – 11): Final Submission

- **Assessment: Final Applied Project Due** (December 9 at 11:59 PM).

Finals Week (Dec 14 – 17)

- No final exam. The Final Applied Project serves as the culminating assessment.